



Church Office Management System

Setup Guide — How to Load and Run This Project on Another
Computer From a ZIP File

Node.js + Express

Angular

MySQL 8

Prepared as a companion document for the project ZIP archive

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Before you start

This project is source code, not a finished installer — the person receiving the ZIP needs a few free tools installed once, then two commands to start it. Nothing here needs an internet connection except the one-time tool downloads and installs below.

Before zipping the project to send it

Delete these folders first — they are large, machine-specific, and get regenerated automatically. Sending them wastes space and can even break things on a different computer (e.g. a different OS/CPU architecture):

- `backend/node_modules/`
- `frontend/node_modules/`
- `frontend/dist/` and `frontend/.angular/`
- `backend/.env` — this holds real secrets and local database credentials that only apply to your computer. The recipient creates their own from `.env.example` (Step 3).

What should be in the ZIP

The `backend/` and `frontend/` source folders, the `docs/` folder, `README.md`, and each project's `package.json` / `.env.example`. That's everything needed — `npm install` rebuilds everything else fresh on the new machine.

2 Prerequisites to install on the new computer

Install these once. All are free, and installers are available for Windows, macOS, and Linux.

Tool	Version	Get it from
Node.js (includes npm)	18.x or newer (22.x used in development)	nodejs.org — download the LTS installer and run it
MySQL Server	8.0	dev.mysql.com/downloads/installer/ (Windows) or brew install mysql (macOS) / your Linux package manager
A way to unzip	—	Built into Windows/macOS; Linux: unzip

✓ How to check what's already installed

Open a terminal (Command Prompt / PowerShell / Terminal) and run:

```
node -v
npm -v
mysql --version
```

If all three print a version number, that tool is already installed and this step can be skipped for it.

No Angular CLI install is needed — the commands below use [npx](#) / [npm](#) scripts that fetch it automatically. No paid license, font, or service is required to run any part of this app.

3 Step 1 — Extract the ZIP

Right-click the ZIP file → **Extract All** (Windows) or double-click it (macOS) → choose a simple destination folder, e.g. `C:\coms` or `~/coms`. Avoid paths with unusual characters or very long nesting.

After extracting, you should see two folders side by side: `backend/` and `frontend/`.

4 Step 2 — Set up MySQL

Make sure the MySQL service is running (the installer starts it automatically as a background service on Windows/macOS). You only need a MySQL user that can create databases — the default `root` user set up during MySQL's installation is fine.

The app's setup script **creates the database and all tables for you** — there is no separate database-creation step. Just make sure you know the root password you chose while installing MySQL; you'll put it in a config file in the next step.

5 Step 3 — Backend (the API)

Open a terminal inside the extracted project folder and run:

```
# move into the backend folder
cd backend

# create your local config file from the template
cp .env.example .env
```

Open the new `backend/.env` file in any text editor and set your MySQL credentials:

```
DB_HOST=127.0.0.1
DB_PORT=3306
DB_USER=root
DB_PASSWORD=<the MySQL root password you set during install>
DB_NAME=coms_db
```

⚠ Replace the JWT secrets

While editing `.env`, also replace `JWT_ACCESS_SECRET` and `JWT_REFRESH_SECRET` with your own random values — never reuse the placeholder text. Generate two quick random strings with:

```
node -e "console.log(require('crypto').randomBytes(48).toString('hex'))"
```

Run it twice and paste one result into each secret.

Now install and start the backend:

```
# installs dependencies AND extracts the PDF fonts automatically
npm install

# creates the database + tables, then seeds starter data (safe to re-run)
npm run db:setup

# starts the API server
npm run dev
```

On success, the terminal prints the API address (`http://localhost:4000`) and, the first time only, the seeded admin login:

```
Admin login created -> username: admin password: Admin@12345
You will be required to change this password on first login.
```

Leave this terminal window open — closing it stops the API server.

6 Step 4 — Frontend (the website)

Open a **second**, separate terminal window (keep the backend one running) and run:

```
# move into the frontend folder
cd frontend

# installs dependencies
npm install

# starts the website
npm start
```

This starts the website at <http://localhost:4200> and automatically forwards its API calls to the backend on port 4000. Leave this second terminal open too.

7 Step 5 — First login

Open a web browser and go to:

```
http://localhost:4200
```

Username	admin
Password	Admin@12345

You'll be forced to set a new password immediately on first login. After that:

- **Masters** — review/edit the seeded church details, Mass schedule, and intention list to match the actual parish.
- **Settings → User Management** — create real staff accounts (each one gets an auto-generated temporary password shown once).
- **Settings → Roles & Permissions** — adjust what each role can do.

8 Letting other computers on the network use it (optional)

By default the app only answers on the computer it's running on (`localhost`). To let other computers/phones on the same office network reach it from the machine's own network (LAN) address:

1. Find that computer's local IP address (Windows: `ipconfig` ; macOS/Linux: `ifconfig` or `ip a`) — e.g. `192.168.1.20` .
2. In `backend/.env` , add that address to `CORS_ORIGIN` and set `PUBLIC_APP_URL` to it, e.g.:

```
CORS_ORIGIN=http://localhost:4200,http://192.168.1.20:4200
PUBLIC_APP_URL=http://192.168.1.20:4200
```

3. Restart the backend (`Ctrl+C` then `npm run dev` again).
4. Other devices on the same Wi-Fi/network can then open `http://192.168.1.20:4200` .

⚠ Note on fingerprint/passkey sign-in

Browsers only allow biometric/passkey sign-in over HTTPS, or over plain HTTP on `localhost` specifically — it will not work over a plain-HTTP LAN address like the one above. Password sign-in works everywhere regardless.

9 Troubleshooting

Symptom	Likely cause / fix
Website shows "Cannot reach the server" / backend terminal shows <code>ECONNREFUSED</code>	MySQL isn't running, or the credentials in <code>backend/.env</code> are wrong. Test with: <code>mysql -h 127.0.0.1 -P 3306 -u root -p</code>
Login succeeds but every page immediately logs back out (401 errors)	The browser is blocking the login cookie because <code>CORS_ORIGIN</code> in <code>.env</code> doesn't exactly match the URL used to open the site (including the port number). Make them match exactly, then restart the backend.
PDFs (receipts / register / certificates) come back empty or error out	Run <code>npm run fonts:extract</code> inside <code>backend/</code> to regenerate the bundled font files — no download needed, it extracts from a package already installed.
<code>npm install</code> fails or hangs	Check the internet connection (npm needs it to download packages the first time), or re-run the command — partial installs resume safely.
Port 4000 or 4200 "already in use"	Another program is using that port. Close it, or change <code>PORT</code> in <code>backend/.env</code> (and update the frontend's proxy config to match).

10 Quick command reference

Once everything is installed, this is all it takes to start the app again later (e.g. the next day, after restarting the computer):

```
# Terminal 1
cd backend
npm run dev

# Terminal 2
cd frontend
npm start

# Then open
http://localhost:4200
```

No need to repeat `npm install` or `npm run db:setup` after the very first time, unless dependencies change or the database is reset.

Church Office Management System — Setup Guide · For the full reference, see docs/INSTALLATION.md and docs/API.md included in the project